



Weekly report 1

COURSE

Master Thesis

SUPERVISOR

Rasmus R. Paulsen
Paula Lopez Diez
Esther Creixell Mediente

AUTHOR

Aïda Jiménez Ordoñez

JANUARY 22, 2026

1 Week 1,2: 11-23. January 2026

1.1 Project Plan

A preliminary project schedule is provided in Table 1. This plan remains flexible, with any required modifications to the timeline being managed through close collaboration with my supervisors.

Week	Activity	Risk
1	Project kick-off	1
2-4	Literature review: ear anatomy, AI-driven segmentation models, acoustic and mechanical feedback, and the finite element method (FEM)	1
2-4	Data acquisition and analysis: become familiar with ear CT scans and analyse data demographics and image quality	2
5-6	Data pre-processing and AI-driven model generation for soft-tissue segmentation	3
7-10	AI-driven model generation for segmentation of different structures: bone, cartilage, air, and skin (if not possible, segmentation based on thresholding and manual annotation)	5
11-14	Geometry processing and mesh generation of biological structures: conversion of ear and surrounding tissue segmentation results into geometries suitable for FEM simulations	4
13-16	Geometry processing and mesh generation of the hearing aid: selection of the appropriate hearing aid, generation of the correct geometry, and placement of the hearing aid in the correct position within the biological structure	5
17-21	Vibro-acoustic simulations: preparation and execution of mechanical simulations of vibro-acoustic feedback using Ansys Mechanical	5
21-24	Visualization and analysis of the results	3
24	Report submission	2
26	Oral presentation	1

Table 1: Project plan for the development of my thesis. Each activity has an associated risk of delay ranging from 1 (low risk) to 5 (high risk).

1.2 Master Thesis Objectives

The overall goal of this MSc thesis project is to develop high-fidelity three-dimensional anatomical models of the human head and ear based on medical imaging data. The project encompasses the following main objectives:

- Understand the anatomical structure of the head and ear, with particular emphasis on tissues relevant for vibroacoustic behavior.

- Process and analyze three-dimensional medical imaging data for anatomical modeling.
- Investigate tissue characterization techniques for realistic representation of ear and head anatomy.
- Develop, or partially develop, an AI-based pipeline for the automated generation of three-dimensional anatomical models from raw medical imaging data.
- Generate high-quality surface and volumetric meshes suitable for numerical simulations.
- Adapt the anatomical models for vibroacoustic simulations of the ear, including mesh optimization and model refinement.
- Set up and execute vibroacoustic simulations using appropriate boundary conditions and material properties.
- Perform comparative simulation studies across different anatomical variations.
- Analyze the influence of tissue properties, anatomical diversity, and boundary condition configurations on simulation results.
- Critically review relevant scientific literature and relate state-of-the-art methods to the results obtained in this thesis.

Throughout the project, continuous evaluation of the proposed methods will be performed, and the outcomes will be compared with existing approaches in order to ensure scientific validity and practical relevance.

A preliminary flowchart of the proposed pipeline for the project is shown in Figure 1.

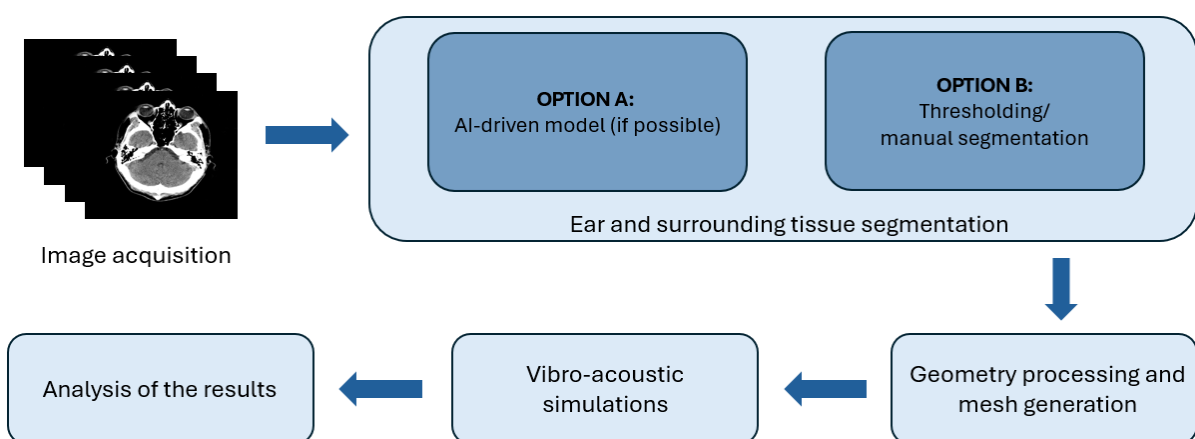


Figure 1: Preliminary pipeline for the project.

1.3 Problem Statement

Hearing loss represents a global health problem, currently impacting the quality of life for approximately 1.5 billion people worldwide [1]. While hearing aids are the primary solution to this challenge, their performance is frequently reduced by vibroacoustic feedback. This phenomenon occurs when mechanical vibrations from the receiver are transmitted through the device and the surrounding ear tissue back to the microphone, resulting in unstable oscillations and audible whistling. Effectively mitigating this feedback requires precise computational models that can simulate how vibrations propagate through the specific biological structures of the ear. However, a major obstacle in developing these models is the limited availability of high-quality anatomical data [1, 2, 3, 4].

Traditionally, researchers have relied on silicone impressions to capture ear geometry, but this method introduces significant physical inconsistencies that skew feedback predictions. For instance, the measured length of the ear canal is often affected by the insertion depth of a cotton block, and the boundary of the external ear surface is determined by the specific volume of silicone used, leading to the inconsistent placement of ear canal inserts within simulations [1]. These geometric inaccuracies directly impact the modeled coupling between the hearing aid and the ear, making it difficult to isolate how much feedback is caused by the device itself.

To solve these limitations, simulation techniques have evolved to better represent how ear tissues affect vibroacoustic behavior. Older simulations contained only the ear canal and used different geometry layers for skin and cartilage, although the skin layer was modeled with material properties that were too soft to provide realistic mechanical resistance. Newer simulations have expanded this scope to include the pinna and assign tissue properties that more closely represent the actual stiffness of human cartilage and skin. This shift is critical because the vibration of a hearing instrument is highly dependent on the mechanical impedance and damping characteristics of the surrounding ear canal tissue [5, 6].

Critical challenges remain in the effort to characterize the surrounding tissue of the ear, automating the generation of individual tissue layers from medical scans. Additionally, the simulation needs to include inter-subject variability since the impact of biological tissue effects varies from person to person. Addressing all these issues ensures that a better accurate-based approach for the design of more effective feedback-free hearing aid devices can be developed.

1.4 Biological background

1.4.1 Ear anatomy

To create more effective models, it is essential to understand the complex anatomy of the ear, which is divided into the inner, middle, and outer ear [7] (see Figure 2). The middle ear is the responsible to transform the sound pressure captured at the tympanic membrane into mechanical vibrations within the ossicular chain, playing essential role in sound transmission. The tympanic membrane itself is a thin, conical, multilayered struc-

ture that separates the ear canal from the middle ear. It acts as both an acoustic receiver and a mechanical interface.

The outer ear consists of the pinna and the concha, which help collect and direct sound into the ear canal. This canal extends from the base of the concha to the tympanic membrane, measuring between 20 and 37 millimeters in length with a diameter ranging from 4.5 to 11 millimeters. Most ear canals feature two bends, with the transition between them known as the isthmus (Figure 3).

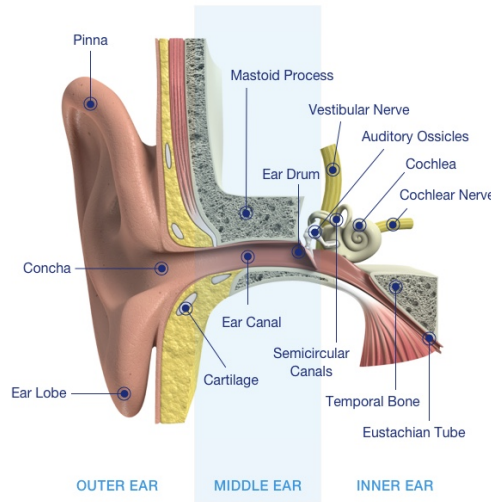


Figure 2: Ear canal anatomy [8].

While the pinna is composed of cartilage, the canal itself is heterogeneous: approximately two-thirds of the lateral part is cartilaginous and soft, covered by skin 1 millimeter thick, while the remaining medial third is osseous, forming a hard structure covered by skin only 0.1 millimeters thick (Figure 3). The meeting point of these sections is the cartilage-bone junction (CBJ). Identifying the CBJ is critical because it typically occurs after the second bend, and for accurate hearing aid fittings, impressions must extend 3 to 5 millimeters beyond this second bend to ensure the junction is included [4].

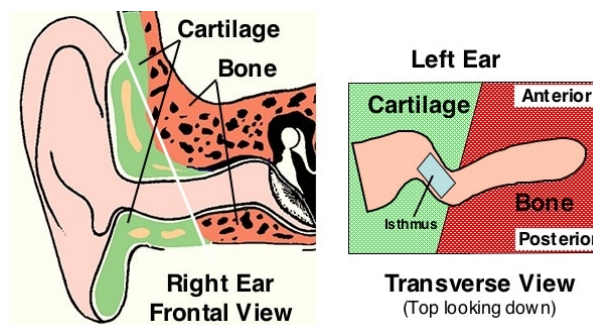


Figure 3: Bone-cartilage canal anatomy [9].

1.4.2 Bone conduction and the occlusion effect

Bone conduction refers to hearing mechanisms whereby skull vibrations lead to auditory perception through multiple transmission pathways [10, 11]. One important phenomenon in this context is the occlusion effect, characterized by a substantial increase in low-frequency sound pressure in the ear canal when it is occluded [12, 13].

The importance of identifying the bony portion lies in its role in managing this occlusion effect. This phenomenon is caused by vibrations of the walls in the soft, cartilaginous part of the ear canal, which generate unwanted sound pressure that users perceive as a "boomy" or "hollow" sound. A key solution is to design a hearing aid that fits with a secure seal in the bony portion of the ear canal, which prevents these vibrations in the soft tissue from being trapped and amplified. However, if this seal is broken or poorly fitted, the occlusion effect increases significantly. Therefore, it is important to ensure the hearing instrument maintains a precise bony seal, which again underscores the need for accurate anatomical identification of the osseous region during the modeling and fitting process.

The occlusion effect typically manifests as a 20–30 dB increase at low frequencies and is particularly noticeable when a person speaks or chews while wearing earplugs or hearing aids with closed fittings [14, 15]. While the effect is useful in certain clinical diagnostics, it can significantly reduce comfort for hearing aid and hearing protection users [16].

Experimental evidence has established that the occlusion effect at low frequencies is primarily driven by cartilage-bone vibrations in the ear canal walls [13, 16]. While other transmission paths, such as the mandible or tympanic membrane compliance, exist, they appear to play a secondary role. Measurements further demonstrate that the magnitude of the occlusion effect is heavily dependent on factors such as insertion depth, occluding volume, material stiffness of the earplug, and individual anatomy [15, 5]. These findings highlight the necessity for more detailed modeling approaches capable of capturing three-dimensional geometry and material-specific phenomena.

1.4.3 Role of ear anatomy in vibro-acoustics

The transmission of vibrations depends significantly on ear anatomy [17]. The geometry of the ear canal, pinna, skin, cartilage, and temporal bone influences the dynamic response of the system. Different hearing aid styles, such as in-the-ear and receiver-in-the-ear devices, interact with these anatomical structures in distinct ways.

Numerical simulations of vibroacoustic feedback therefore require realistic three-dimensional ear geometries, appropriate tissue layering, material properties, and boundary conditions. Simplified models risk misrepresenting the dominant transmission paths and feedback mechanisms.

1.5 Technical background

1.5.1 Hearing Aid Architecture

Modern hearing aids aim to provide high acoustic gain, compact size, and user comfort. From a physical perspective, their performance is fundamentally limited by feedback. Feedback occurs when sound or vibration produced by the receiver re-enters the microphone path, creating a closed loop that may lead to instability or limit the maximum usable gain.

A hearing aid typically consists of microphones, a receiver, a signal processing unit, and an earpiece or shell that couples the device to the ear canal. Feedback paths may be acoustic, mechanical, or vibroacoustic.

From control theory, the open-loop gain A_{OL} must satisfy

$$A_{OL}\beta \leq 1,$$

where β represents the total feedback transfer function. Reducing β directly increases the stable maximum gain.

1.5.2 Acoustic feedback

Acoustic feedback arises from sound leakage through air paths, typically via vents or imperfect sealing between the earpiece and the ear canal. Venting is often intentionally introduced to mitigate the occlusion effect and improve comfort, but it simultaneously increases acoustic feedback. Signal processing algorithms can partially compensate for acoustic feedback, although they may introduce audible artifacts or reduce effective bandwidth.

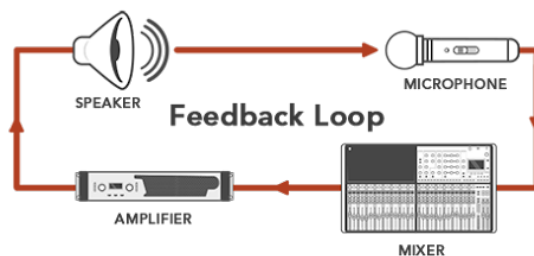


Figure 4: Feedback scheme [18].

Mechanical design therefore remains essential to limit acoustic leakage through appropriate shell geometry, dome selection, and material choice, all of which depend strongly on individual ear canal anatomy.

1.5.3 Vibro-acoustic and mechanical feedback

In addition to purely acoustic paths, vibro-acoustic feedback involves structural vibrations of the receiver, hearing aid housing, earpiece, and surrounding ear tissue. These vibrations can radiate sound into the ear canal and be detected by the microphones.

Unlike purely acoustic feedback, vibro-acoustic feedback is more difficult to suppress using digital algorithms alone. As a result, mechanical design choices such as housing stiffness, mounting of the receiver, damping treatments, and coupling to the ear are critical. Vibro-acoustic feedback is often assessed under test conditions that minimize acoustic feedback, for example by using couplers or specialized fixtures.

1.5.4 AI-driven segmentation models

Medical image segmentation is the process of partitioning a digital medical image into multiple segments or sets of pixels, typically to locate tumors, organs, or other substructures [19]. This critical step allows clinicians to isolate a specific Region of Interest (ROI) for more accurate diagnosis and surgical treatment planning.

Early traditional medical image segmentation methods, such as edge detection, template matching, region growing, and graph cutting, have been used extensively in clinical practice. However, these techniques face significant challenges due to the immense variety of anatomical shapes and sizes found in human patients [19]. Because these methods often rely on fixed rules, they lack the flexibility required for complex cases, highlighting the urgent need to develop more general and adaptive algorithms.

Artificial Intelligence (AI) has emerged as the primary solution to these challenges, providing machines with the ability to learn complex patterns directly from medical data. As detailed in the review of AI methods for medical segmentation by Teng et al. (2024) [19], the field has transitioned from basic automated tools to sophisticated neural networks. Classical AI methods, such as K-means clustering, which simplifies the detection of ROI, Decision Trees, and Support Vector Machines, provided the initial framework for moving beyond manual segmentation.

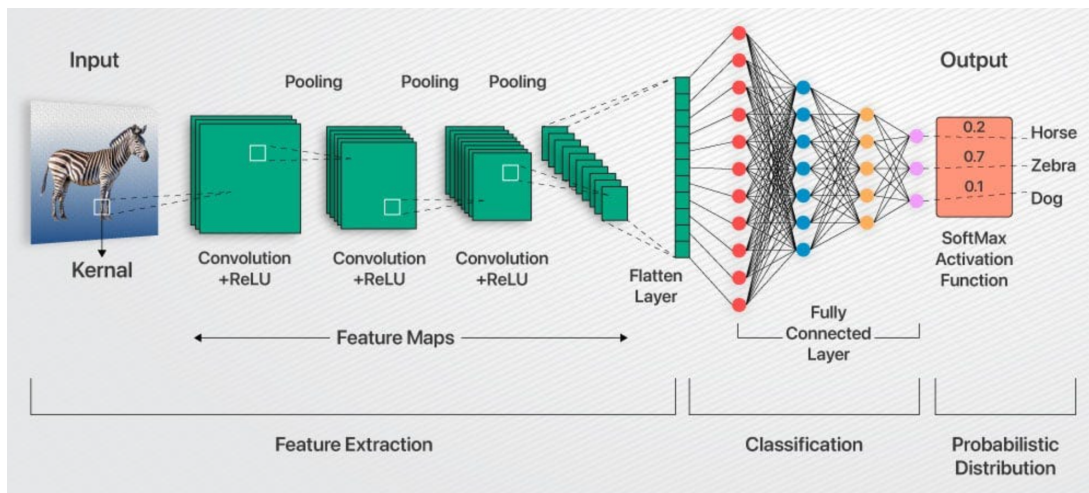


Figure 5: Feedback scheme [20].

Deep learning represents the most advanced tier of AI in this field, with Convolutional Neural Networks (CNNs) leading the way in analyzing images and videos. A standard CNN operates through a structured sequence (see Figure 5): the convolutional layer extracts features, the activation function introduces non-linearity, the pooling layer reduces

data dimensionality, and the fully connected and output layers provide the final segmentation results.

To further enhance these networks, specialized architectures have been developed. ResNets utilize residual learning by introducing skip connections, which allow the model to train much deeper without losing information. Generative Adversarial Networks (GANs) take a different approach, using a generator and a discriminator in a competitive process to create highly realistic segmentation maps.

Finally, one of the most effective modern architectures is TransUNet. This model combines the power of Transformers, which are excellent at understanding the global context of an image, with the U-Net architecture, which is famous for its precise pixel-level localization. By merging these two approaches, TransUNet can handle the most difficult segmentation tasks with high accuracy.

As AI models become more complex, the industry has shifted its focus toward Explainable AI (XAI) and Trustworthy AI (TAI). XAI aims to make the decision-making process of these "black box" models more transparent to doctors, ensuring they understand why a specific diagnosis was made. However, there is still a notable lack of security and robustness in many current models, which TAI seeks to resolve by ensuring that AI systems are reliable, ethical, and safe for high-stakes medical environments.

1.5.5 Finite Element Method

The Finite Element Method (FEM) is a sophisticated numerical technique used to find approximate solutions to complex engineering and mathematical problems by dividing a large, intricate system into smaller, simpler parts called finite elements. This process, known as discretization or meshing, creates a grid of points (nodes) that allows computers to apply differential equations to each element individually rather than attempting to solve for the entire complex geometry at once. In simulations, often referred to as Finite Element Analysis (FEA), FEM is used to predict how physical objects will react to real-world forces such as heat, vibration, fluid flow, and mechanical stress [21, 22].

An scheme of the main FEM steps is shown in Figure 6.

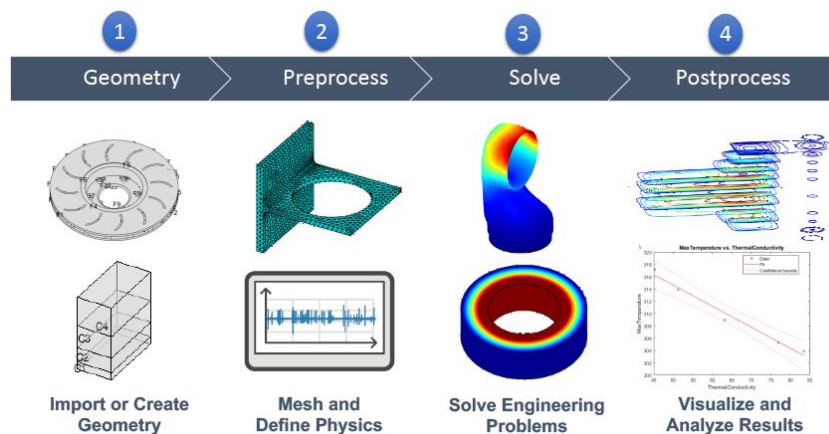


Figure 6: FEM main steps [23].

1.6 State-of-the-art

1.6.1 Anatomical characterization and geometry of the human ear canal

The accurate prediction of sound-pressure level distribution relies fundamentally on the geometric representation of the ear canal. Early foundational work by Stinson (1989) [24] established that a uniform tube approximation of the ear canal is inadequate for acoustic calculations above 8 kHz, necessitating models that include specific cross-sectional areas and longitudinal curvature.

Nielsen and Darkner (2011) [4] advanced the clinical understanding of these geometries by identifying that the CBJ typically lies 1.81 mm medial to the second bend, implying that hearing aid impressions must extend 3 mm to 5 mm beyond this second bend to ensure a proper "bony seal" and mitigate the occlusion effect.

Abdala and Keefe (2012) [25] further contributed by reviewing the maturation of the canal from birth through puberty, identifying how changes in the central axis and entrance morphology influence audiological measurements throughout development. Building on these developmental insights, Voss et al. (2020) [2] utilized silicone molds from 169 live subjects to demonstrate that canal areas are systematically larger than previously assumed, particularly in older adults, and that area typically increases with each decade of life.

Recent longitudinal studies have significantly expanded the understanding of these geometries across the human lifespan. Balouch et al. (2023) [6] further quantified this variability using high-resolution CT scans, finding that canal areas can vary by factors of two to three between individuals at the same anatomical location. Most recently, Voss et al. (2025) [3] provided a comprehensive analysis using CT scans of 221 ears, showing that while the canal becomes fully cartilaginous at the second bend by 0.7 years of age, adult-like geometric characteristics do not fully emerge until between ages 10 and 15.

These findings emphasize that precision in modern audiological diagnostics and deep-canal hearing aid fittings relies on accounting for these complex, age-related structural variations rather than assuming a universal ear canal geometry.

1.6.2 AI-driven models for ear segmentation

The evolution of medical image segmentation for ear and head structures has progressed from semi-automated geometric modeling to state-of-the-art deep learning frameworks.

Poznyakovskiy et al. (2011) [26] introduced an early specialized method for modeling the fluid chambers of the cochlea. Their approach utilized dynamic resampling of tomography image stacks to ensure perpendicular cross-sections, which were then segmented using 2D active contours while a Kalman filter predicted the centerline trajectory.

More recent research has shifted toward fully automated neural networks to improve both speed and precision. Hussain et al. (2021) [27] developed AutoCasNet, a framework designed to segment inner ear structures from micro-CT images. This model outperformed previous algorithms by achieving high-speed processing without sacrificing accuracy. Sim-

ilarly, Neves et al. (2021) [28] implemented a deep learning pipeline using CNNs to segment the inner ear, ossicles, facial nerve, and sigmoid sinus from clinical CT scans. Their system achieved expert-level accuracy while delivering a transformative 90-fold reduction in processing time compared to manual methods.

Advancements have also been made in MRI-based analysis. Vaidyanathan et al. (2021) [29] utilized a 3D U-Net with attention gates to automate inner ear segmentation on MRI scans. Trained on a massive dataset of 944 scans, the model achieved a mean Dice Similarity Coefficient of 0.8790 on independent, multicentric test sets, proving to be as reliable as human readers. Extending the capabilities of the 3D U-Net, Stebani et al. (2023) [30] developed a pipeline that jointly generates segmentation maps and localizes key anatomical landmarks from temporal bone CT scans, demonstrating high generalizability across both in-house and open-source datasets.

Beyond internal structures, AI is now being applied to the outer ear for prosthetic design. Dritsas et al. (2024) [1] presented a CNN-based method to pre-process ear canal impressions. By automating handedness classification, geometry orientation, and artifact removal, they streamlined the manufacturing and fitting of hearing aids. Their evaluation on a large dataset of 3,000 ear canal impressions reached an impressive intersection over union of over 93.4%, facilitating more precise morphological studies of the outer ear canal.

However, characterization of the surrounding tissue of the ear canal is missing.

1.6.3 Classical and lumped-element modeling of the occlusion effect

The occlusion effect, defined as the amplification of bone-conducted sound when the ear canal is blocked, has been modeled through various paradigms since the mid-20th century. Huizing et al. (1960) [31] initially described the phenomenon by comparing the resonance characteristics of open and closed pipes. This was later refined by Tonndorf et al. (1966) [32], who proposed an electrical analog model where the open ear canal acts as a high-pass filter; occlusion removes this filtering effect, thereby trapping and amplifying low-frequency energy. While these classical models explain general trends, they rely on simplified one-dimensional representations of the ear canal and surrounding tissues [14].

To improve physical intuition while maintaining computational efficiency, several authors adopted lumped-element modeling (LEM) approaches [13]. These models represent acoustic and mechanical components through combinations of masses, springs, and resistive elements. Although LEMs can reproduce experimental trends with relatively few parameters, they are inherently limited in spatial resolution. They offer only an indirect correspondence between model parameters and actual anatomical properties, restricting their ability to represent complex geometries and heterogeneous tissues [5].

1.6.4 Three-Dimensional Finite Element Modeling

To overcome the limitations of simplified models, three-dimensional finite element models of the external ear have been developed to explicitly represent ear canal geometry, surrounding tissues, and acoustic-structure interaction. Building on these developments, Brummund et al. (2014) [5], reconstructed the ear anatomy from anatomical image data,

where structural domains like bone, cartilage, and skin were modeled as linear elastic materials. The air within the ear canal was represented as an acoustic domain coupled to these surrounding structures, allowing for the simulation of occlusion by inserting deformable earplug models at varying depths.

The FE model utilized mastoid excitation to represent bone-conducted input, which is consistent with experimental measurement procedures [11]. Boundary conditions were set to approximate the coupling to the surrounding head and the radiation at the ear canal entrance, while the tympanic membrane was represented using an impedance boundary derived from middle-ear models. Simulation results indicated that occlusion generally increases low-frequency sound pressure, with shallow occlusion producing the largest enhancement. Power balance analyses suggest that the effect is not solely due to blocked radiation but rather because occlusion modifies the acoustic impedance seen by the vibrating canal walls, thereby increasing the power transferred into the cavity.

More recently, Elghanaoui et al. (2025) [33] developed and validated a three-dimensional finite element model of the human middle ear derived from high-resolution computed tomography (CT) data. This methodology allowed for the reconstruction of a complete and anatomically coherent model, including the external auditory canal, a layered tympanic membrane, and the ossicular chain, offering a more realistic tool for mechanical simulation compared to previous histological or simplified techniques. By integrating precise geometry into a COMSOL Multiphysics framework, the researchers demonstrated that their model accurately reproduces middle ear dynamics across a wide frequency range. Quantitative validation against experimental benchmarks showed that this CT-based approach achieves lower error metrics (MAE and RMSE) than prior numerical models, particularly within the clinically critical resonance range of 700–2000 Hz, thereby providing a robust framework for auditory diagnostics and the design of biomedical prostheses.

1.7 Research Questions

1. How to draw the tissue layers?
2. Frequency-dependent material properties?
3. Where to cut the ear/head geometry?
4. Boundary conditions?

1.8 Software to use

- 3D Slicer [34] for segmentation.
- Ansys Mechanical [?] for simulation.

References

- [1] Stylianos Dritsas, Kenneth Wei De Chua, Zhi Hwee Goh, and Robert E. Simpson. Classification, registration and segmentation of ear canal impressions using convolutional neural networks. *Medical Image Analysis*, 94:103152, May 2024.
- [2] Susan E. Voss, Nicholas J. Horton, Katherine E. Fairbank, Lu Xia, Lauren R. K. Tinglin, and Kathryn D. Girardin. Measurements of ear-canal cross-sectional areas from live human ears with implications for wideband acoustic immittance measurements. *The Journal of the Acoustical Society of America*, 148(5):3042–3051, November 2020.
- [3] Susan E Voss, Aaron K Remenschneider, Rebecca M Farrar, Soomin Myoung, and Nicholas J Horton. Comprehensive measurements and analyses of ear canal geometry from late infancy through late adulthood: Age-related variations and implications for basic science and audiological measurements. *Trends in Hearing*, 29, June 2025.
- [4] Claus Nielsen and Sune Darkner. The cartilage bone junction and its implication for deep canal hearing instrument fittings. *The Hearing Journal*, 64(3):35, March 2011.
- [5] Martin K. Brummund, Franck Sgard, Yvan Petit, and Frédéric Laville. Three-dimensional finite element modeling of the human external ear: Simulation study of the bone conduction occlusion effect. *The Journal of the Acoustical Society of America*, 135(3):1433–1444, March 2014.
- [6] Auden P. Balouch, Karen Bekhazi, Hannah E. Durkee, Rebecca M. Farrar, Mealakthey Sok, Douglas H. Keefe, Aaron K. Remenschneider, Nicholas J. Horton, and Susan E. Voss. Measurements of ear-canal geometry from high-resolution ct scans of human adult ears. *Hearing Research*, 434:108782, July 2023.
- [7] Huibin Shi, Shuyi Xiang, Liang Wang, Yongtao Sun, Jie Wang, and Zhanli Liu. Characterization of middle ear soft tissue damping and its role in sound transmission. *Biomechanics and Modeling in Mechanobiology*, 22(3):1003–1018, March 2023.
- [8] Hearingexplained-howdowehear?(2024)BayAudio.Availableat:<https://www.bayaudio.com.au/how-do-we-hear/>(Accessed:12January2026)..
- [9] Wayne Staab. The Human Ear Canal: Anatomy and Physiology | Wayne Staab, PhD — hearinghealthmatters.org. <https://hearinghealthmatters.org/waynesworld/2023/human-ear-canal/>.
- [10] J. Tonndorf. The mechanism of bone conduction. *Annals of Otology, Rhinology & Laryngology*, 75:101–123, 1966.
- [11] S. Stenfelt. Bone-conducted sound: Physiological and clinical aspects. *Audiology and Neurotology*, 10:193–211, 2005.
- [12] E. H. Huizing. Bone conduction: The influence of the middle ear. *Acta Otolaryngologica Supplementum*, 155:1–99, 1960.

- [13] S. Stenfelt and S. Reinfeldt. A model of the occlusion effect with bone-conducted stimulation. *International Journal of Audiology*, 46:595–608, 2007.
- [14] J. Schroeter and U. Poesselt. The occlusion effect and its influence on bone conduction hearing aids. *Journal of the Acoustical Society of America*, 80:128–132, 1986.
- [15] M. Hansen. *Occlusion effects, Part II: A study of the occlusion effect mechanism and the influence of the earmould properties*. PhD thesis, Technical University of Denmark, 1998.
- [16] S. Stenfelt. Acoustic and physiologic aspects of bone conduction hearing. *Advances in Oto-Rhino-Laryngology*, 71:10–21, 2010.
- [17] Martin K. Brummund, Franck Sgard, Yvan Petit, and Frédéric Laville. On the influence of the material properties of the external ear on occlusion effect simulations.
- [18] How to Control Feedback — news.harman.com. <https://news.harman.com/blog/how-to-control-feedback>.
- [19] Zixuan Teng, Lan Li, Ziqing Xin, Dehui Xiang, Jiang Huang, Hailing Zhou, Fei Shi, Weifang Zhu, Jing Cai, Tao Peng, and Xinjian Chen. A literature review of artificial intelligence (ai) for medical image segmentation: from ai and explainable ai to trustworthy ai. *Quantitative Imaging in Medicine and Surgery*, 14(12):9620–9652, December 2024.
- [20] Ravjot Singh. Decoding cnns: A beginner’s guide to convolutional neural networks and their applications.
- [21] Detailed Explanation of the Finite Element Method (FEM) — comsol.com. <https://www.comsol.com/multiphysics/finite-element-method>.
- [22] What Is FEM & FEA Explained | Finite Element Method | SimScale — simscale.com. <https://www.simscale.com/blog/what-is-finite-element-method/>.
- [23] <https://de.mathworks.com/discovery/finite-element-analysis.html>.
- [24] Michael R. Stinson and B. W. Lawton. Specification of the geometry of the human ear canal for the prediction of sound-pressure level distribution. *The Journal of the Acoustical Society of America*, 85(6):2492–2503, June 1989.
- [25] Carolina Abdala and Douglas H. Keefe. *Morphological and Functional Ear Development*, page 19–59. Springer New York, November 2011.
- [26] Anton A. Poznyakovskiy, Thomas Zahnert, Yannis Kalaidzidis, Nikoloz Lazurashvili, Rolf Schmidt, Hans-Jürgen Hardtke, Björn Fischer, and Yury M. Yarin. A segmentation method to obtain a complete geometry model of the hearing organ. *Hearing Research*, 282(1–2):25–34, December 2011.
- [27] Raabid Hussain, Alain Lalande, Kibrom Berihu Girum, Caroline Guigou, and Alexis Bozorg Grayeli. Automatic segmentation of inner ear on ct-scan using auto-context convolutional neural network. *Scientific Reports*, 11(1), February 2021.

- [28] C. A. Neves, E. D. Tran, I. M. Kessler, and N. H. Blevins. Fully automated preoperative segmentation of temporal bone structures from clinical ct scans. *Scientific Reports*, 11(1), January 2021.
- [29] Akshayaa Vaidyanathan, Marly F. J. A. van der Lubbe, Ralph T. H. Leijenaar, Marc van Hoof, Fadila Zerka, Benjamin Miraglio, Sergey Primakov, Alida A. Postma, Tjasse D. Brintjes, Monique A. L. Bilderbeek, Hammer Sebastiaan, Patrick F. M. Dammeijer, Vincent van Rompaey, Henry C. Woodruff, Wim Vos, Seán Walsh, Raymond van de Berg, and Philippe Lambin. Deep learning for the fully automated segmentation of the inner ear on mri. *Scientific Reports*, 11(1), February 2021.
- [30] Jannik Stebani, Martin Blaimer, Simon Zabler, Tilmann Neun, Daniël M. Pelt, and Kristen Rak. Towards fully automated inner ear analysis with deep-learning-based joint segmentation and landmark detection framework. *Scientific Reports*, 13(1), November 2023.
- [31] David Chhan, Christof Rösli, Melissa L. McKinnon, and John J. Rosowski. Evidence of inner ear contribution in bone conduction in chinchilla. *Hearing Research*, 301:66–71, July 2013.
- [32] D.R.F Irvine and K.G Wester. Bone conduction as a means of acoustic input control: the effects of middle ear muscle contractions. *Electroencephalography and Clinical Neurophysiology*, 34(1):80–82, January 1973.
- [33] Souad Elghanaoui, Adil Faiz, Safa Assif, and Abdelwahed Hajjaji. Development and validation of a finite element model of the human middle ear based on medical imaging. *BIO Web of Conferences*, 200:01014, 2025.
- [34] 3D Slicer image computing platform — slicer.org. <https://www.slicer.org/>.